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SVM Documentation

1 Space-Vector Modulation (SVPWM)

Space-Vector PWM (SVPWM) synthesises three-phase voltages from a two-level inverter, maximising DC-bus utilisation and reducing harmonic content relative to sinusoidal PWM.

Rather than treating each phase independently, SVPWM represents the three-phase system as a single vector in the stationary $\alpha\beta$ plane:

$$\vec{V}_{ref} = V_\alpha + jV_\beta$$

The inverter can synthesise eight discrete voltage vectors: six active vectors, forming a hexagon in $\alpha\beta$, and two null vectors at the origin. SVPWM approximates $\vec{V}_{ref}$ by combining, in time, the two active vectors adjacent to it and the null vectors, within every switching period.

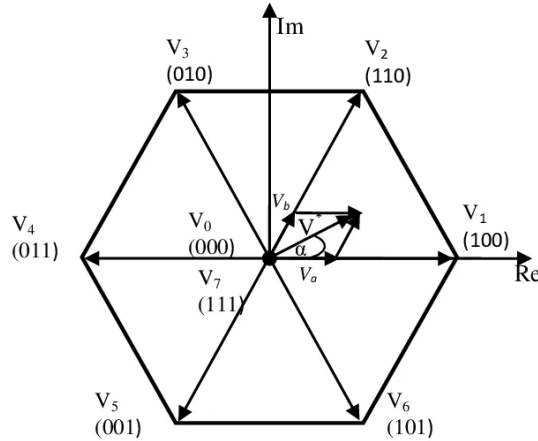


Figure 1: SVPWM hexagon diagram

1.1 Sector and Switching-Time Decomposition

The $\alpha\beta$ plane is split into six 60° sectors:

$$\text{sector} = \left\lfloor \frac{\theta}{\pi/3} \right\rfloor, \quad \theta = \tan^{-1}(V_\beta/V_\alpha) \quad (1)$$

Within each sector, $\vec{V}_{ref}$ is synthesised from the two active vectors adjacent to it:

$$\vec{V}_{ref} = T_1 \vec{V}_1 + T_2 \vec{V}_2, \quad T_1 = V_{norm} \sin\left(\frac{\pi}{3} - \alpha\right), \quad T_2 = V_{norm} \sin(\alpha) \quad (2)$$

where α is the angle measured from the start of the sector and V_{norm} is the reference voltage normalised to the maximum linear value (Section 1.3). $T_1, T_2 \in [0, 1]$ are fractions of the switching period; the remainder is assigned to the null vectors, $T_0 = 1 - T_1 - T_2$.

1.2 Symmetric 7-Segment Sequence and Duty-Cycle Mapping

To reduce harmonic content and keep the switching pattern symmetric, a 7-segment sequence is used, splitting T_0 evenly between the start and end of the period:

$$V_0 \rightarrow V_1 \rightarrow V_2 \rightarrow V_7 \rightarrow V_2 \rightarrow V_1 \rightarrow V_0$$

Defining the cumulative intervals

$$T_a = \frac{T_0}{2}, \quad T_b = T_1 + \frac{T_0}{2}, \quad T_c = T_2 + \frac{T_0}{2}, \quad T_d = T_1 + T_2 + \frac{T_0}{2} \quad (3)$$

the duty cycle of each phase is one of $\{T_a, T_b, T_c, T_d\}$, selected by a permutation table indexed by sector:

Sector	d_a	d_b	d_c
0	T_d	T_c	T_a
1	T_b	T_d	T_a
2	T_a	T_d	T_c
3	T_a	T_b	T_d
4	T_c	T_a	T_d
5	T_d	T_a	T_b

This table is exactly the `TI_Matrix` array in the code (each row a sector, each entry an index into $\{T_a, T_b, T_c, T_d\}$): it avoids conditional branches, guarantees the symmetric 7-segment pattern by construction, and its structure reflects which phase stays connected to the positive bus rail the longest in each sector.

1.3 DC-Bus Utilisation

One of SVPWM's main advantages is better use of the DC-bus voltage. The maximum reference voltage in the linear region is

$$V_{max} = \frac{V_{dc}}{\sqrt{3}} \quad (4)$$

compared to sinusoidal PWM's $V_{max,SPWM} = V_{dc}/2$ — an increase of $2/\sqrt{3} - 1 \approx 15.5\%$ in usable voltage.

2 The SVM() Implementation

The algorithm is implemented as a single function, `SVM()`, taking V_α, V_β and returning duty cycles $d_a, d_b, d_c \in [0, 1]$. It does *not* generate gate edges: those come from a native PLECS triangular carrier and comparators, synchronised so its trough aligns with the control sampling instant (true regular-sampled 7-segment SVPWM).

2.1 Reference Vector

Magnitude and angle are computed directly from V_α, V_β , then θ is wrapped to $[0, 2\pi)$:

$$V = \sqrt{V_\alpha^2 + V_\beta^2}, \quad \theta = \text{atan2}(V_\beta, V_\alpha)$$

2.2 DC-Link Feedforward and the Modulation Margin

Two details are not in the pure theory of Section 1.3, but are needed for a working implementation:

- **Which V_{dc} to normalise with.** The DC-link measurement carries switching and 2ω ripple; dividing by it directly would inject that noise into every duty cycle. If the link is stiff and regulated, the code instead normalises with the nominal bus voltage $V_{dc,nom}$, selected by the flag `use_nominal_vdc`:

$$V_{dc,eff} = \text{use_nominal_vdc} ? V_{dc,nom} : V_{dc}$$

- **Keeping duties off the rails.** V is clamped to $V_{lin} = V_{max}(1 - \epsilon)$, with $\epsilon = \text{SVM_MOD_MARGIN} = 0.02$, *before* normalising by V_{max} :

$$V_{max} = \frac{V_{dc,eff}}{\sqrt{3}}, \quad V \leftarrow \min(V, V_{max}(1 - \epsilon)), \quad V_{norm} = \frac{V}{V_{max}}$$

This keeps $V_{norm} \leq 1 - \epsilon$, so no duty cycle ever reaches exactly 0 or 1 — i.e. a minimum on-time is preserved, matching a real inverter's constraint and avoiding zero-width pulses.

2.3 Sector, Switching Times, and Duty Cycles

With V_{norm} in hand, the sector, switching times and duty-cycle assignment follow Eqs. (1)–(3) of Section 1.1–1.2 exactly, with one addition: the sector index is taken modulo 6 as a safety net against floating-point rounding at $\theta \approx 2\pi$.

$$\text{sector} = \left\lfloor \theta \cdot \frac{1}{\pi/3} \right\rfloor \bmod 6, \quad \alpha = \theta - \text{sector} \cdot \frac{\pi}{3}$$

T_1, T_2 and T_a, T_b, T_c, T_d are computed as in Eqs. (2), (3), and `TI_Matrix` (the table of Section 1.2) picks out d_a, d_b, d_c .

2.4 Output Saturation

Each duty cycle is finally clamped to $[0, 1]$, guaranteeing a realisable reference is always handed to the modulator even under a transient that V_{norm} 's clamp alone does not fully cover.

2.5 Equation $\leftrightarrow$ Code Correspondence

Lines	Code	Equation / concept
26–27	<code>V = sqrtf(...)</code>	vector magnitude
32–34	<code>Theta = atan2f(...) + wrap</code>	vector angle, $[0, 2\pi)$
39	<code>Vdc_eff = ...</code>	DC-link feedforward choice
40–43	<code>V_max, V_lin, V_scaled</code>	Eq. (4) + margin
45–46	<code>sector, alpha</code>	Eq. (1)
48–49	<code>T1, T2</code>	Eq. (2)
51–55	<code>T_intervals[0..3]</code>	T_a, T_b, T_c, T_d , Eq. (3)
57–59	<code>TI_Matrix</code> lookup	duty-cycle table
61–66	clamp to $[0, 1]$	output saturation

SVPWM thus delivers three-phase voltages with low harmonic content and efficient DC-bus utilisation, at the cost of the extra sector logic over plain sinusoidal PWM.

References

- [1] Feshara, H., Elharony, M., & Sharaf, S. (2014). Design, Digital Control, and Simulation of a Grid-Connected Photovoltaic Generation System. *International Journal of Renewable Energy Research*, 4(2), 484-491. Retrieved from ResearchGate. https://www.researchgate.net/publication/275037733_Design_Digital_Control_and_Simulation_of_a_Grid-Connected_Photovoltaic_Generation_System